Rank-27 Congruence ITE Dense-Slot Repair

Tactix Development Project

24 August 2026

Executive summary

This increment promotes a second general representation repair for Congruence ITE extraction. After the accepted ternary sign-mask repair, residual tracing at the exact Rank-27 conflict-9450 common state found that Tactix spent 210 of 415 ITE milliseconds inserting 4,585,536 conditional pairs through an `unordered_map`. That collection alone cost 1.79 times current instrumented Kissat's complete 117 ms ITE stage. Kissat's source-shaped counterpart uses literal-indexed stacks.

The repair maps each dense literal id directly to a sparse owned pair-vector slot. Only touched literals own vectors, and a sparse key list is sorted before consumption exactly as before. Canonical record traversal, per-literal append order, candidate pairing and verification, logical ticks and budgets, scheduling, gates, merges, units, proof transactions, learning, pruning, and N1–N4 are unchanged. No benchmark identity, rank, deadline, arm, or tuned constant enters the solver.

At the exact common state, ITE falls from 572 to 184 ms, complete Congruence from 2383 to 1953 ms, the solve prefix from 16517 to 16147 ms, and peak RSS from 406028 to 394776 KiB. Total sampled cycles fall 3.32%, and the exact 64-row event/order trace remains byte-identical. The complete frozen B28 plus fixed C5 guard passes 66/66 independently checked tasks with zero residue.

Checked Rank-27 job `1646031` proves UNSAT in 399.152 internal / 399.668 wall seconds, versus 409.438 / 409.957 for the accepted predecessor. The 1,514,395,111-byte proof is byte-identical and independently verified before deletion; every non-time statistic is exact. Congruence improves 16.154 to 14.720 seconds and peak solver RSS falls 643976 to 604272 KiB. Rank 27 remains above the 300-second engineering milestone and frozen Kissat's 278.187 seconds, so convergence continues.

The full Tier-D arrays remain canceled and non-scorable. A new repository policy now assigns every future fuzz submission, monitoring pass, artifact inspection, cleanup audit, and terminal report exclusively to Claude Code. Codex will accept Claude's pass/failure report without probing fuzz artifacts; failure returns directly to Claude's bounded repair workflow.

1 Authority and residual localization

The immutable Rank-27 input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Every Rank-27 task uses two CPUs and the pre-Phase-4.5 16 GiB/1200 s contract. Frozen Kissat 4.0.4 and all other frozen references were not rerun. Current instrumented Kissat is source-observer evidence only, never scoring evidence.

Residual observer `1645957` reproduces the accepted conflict-9450 state: 9450 conflicts, 5,795,248 decisions, 96,338,812 propagations, 1,734,747 gates, 58,738 merges, 181 units, 445,868,596 Congruence ticks, and 1,744,090 closure items. Across three ITE calls it visits 1,793,616 canonical records, inserts

4,585,536 conditional pairs under 1,064,350 literal keys, tries 9,335,334 pairs, reaches 2,381,310 complementary-condition trials, and verifies 137,334 definitions. Its timing split is 210 ms collection, 66 ms key preparation and sort, 69 ms pair evaluation, and roughly 70 ms surrounding container lifetime and timer granularity.

The source owner is the map used to route every append and later materialize the key set. The safe repair is representation-only: allocate a `uint32_t` slot image for the dense literal universe, append each newly touched literal to a sparse key list, and allocate pair vectors only for those touched keys. Sorting the key list retains the old canonical traversal; each vector retains insertion order. The two-entries-per-variable image replaces hash buckets and nodes without making physical layout a proof or strategy authority. Final source-diff SHA-256 is `3667828ec6669c3ae5c0dd5a4b8f3e71b469a429a8dddf3e6294dd8846882e83`; Release pin SHA-256 is `60fe8eae742d185e96aac6cd6a94c49bccc3bf23a852464f729e63e2a16ef366`.

2 Exact performance and order evidence

Paired stage `1645958` runs predecessor and candidate simultaneously on node017. After timing and arm labels are removed, the complete logs are byte-identical with SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`. This fixes all printed candidates, work counters, schedule aggregates, learning, pruning, and resulting state.

Measure	Predecessor	Candidate	Ratio
ITE extraction, ms	572	184	0.32168x
Complete Congruence, ms	2383	1953	0.81956x
Inprocessing prefix, ms	7953	7601	0.95574x
Whole solve prefix, ms	16517	16147	0.97760x
Peak RSS, KiB	406028	394776	0.97229x

Paired profile `1645960` reports zero lost samples and again preserves the exact normalized state. Total sampled cycles fall from 65,038,405,342 to 62,875,953,119 (0.966751x), saving 2,162,452,223 cycles. The materialized ITE membership lambda falls 414,600,299 to 163,409,696 cycles; 366,782,892 cycles in predecessor hash-node insertion disappear. `congruence_apply` falls 1,008,906,926 to 803,493,973 cycles. No cost is displaced into another Congruence path.

Event job `1645961` emits exactly 64 untruncated rows with SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`, byte-identical to the frozen accepted trace. This is the direct authority for exact rule and candidate order, logical tick/cost fields, scheduling order, boundary state, learning, and resulting pruning.

3 Correctness and regression gates

Focused Release checks pass 7/7 and focused Debug ASan/UBSan checks pass 5/5. The complete executable Release suite passes 73/73. The opt-in documentation size policy remains known debt and is not an executable or correctness failure. The proof-rule/strategy boundary check passes.

Full-width campaign `benchmarks/runs/20260824-2032-BC-ite-dense-slot-guard/`, array `1645964` and aggregate `1646030`, validates immutable predecessor and candidate arms over all frozen Tier B and the inherited five Congruence-relevant Tier C inputs. Each arm passes 17 SAT models and 16 UNSAT proofs. All 33 non-time solver images are exact. Invalid, missing, regression, cleanup-failure, and certificate-residue counts are zero. Candidate PAR-2 is 1650.796 versus 1630.107

seconds, ratio 1.0126918, below the 1.10 gate. The largest wall increases are exact-state search/Vivify noise outside the mechanism; on T105 the candidate reduces Congruence 33.635 to 30.553 seconds and total solve time 419.293 to 417.822 seconds. Summary SHA-256 is `4675a3f289b68ac983b87c9f8710190dadba3219e6de59dd49dae8466066ac4f`.

Exact-source native-LRAT campaign `1645963`, submitted before the new Claude-owned-fuzzing rule took effect, reports 10,000 random cases and zero failures: 2,624 SAT models, 3,484 UNSAT proofs, 3,892 declared five-second nonterminations with no oracle claim, 3,564 verdict cases reaching Congruence, and 1,253,848 target applications. The 25 retained timeout CNFs are diagnostic inputs only. Future fuzz evidence will be accepted solely from Claude Code's terminal report; Codex will neither submit nor inspect those jobs.

4 Checked Rank-27 result and next actions

Immutable campaign `benchmarks/runs/20260824-2035-tierd-rank27-ite-dense-slot-witness/`, job `1646031`, completes on node017 in 00:08:28. Tactix reports UNSAT in 399.152 internal / 399.668 wall seconds with 604272 KiB peak solver RSS. Pinned `drat-trim` checks the original CNF and the 1,514,395,111-byte proof, reports `s VERIFIED`, and exits zero in 106.719 internal / 106.878 wall seconds. Proof SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d` is byte-identical to the predecessor. Proof, input, checker derivatives, and scratch are deleted before atomic publication; certificate residue is zero.

Removing timing lines makes every solver statistic byte-identical to predecessor `1645956`, normalized SHA-256 `8cd3eda481799871a179e1483a57413622bc7832c09432df705b2308bafc2da7`. Congruence improves 16.154 to 14.720 seconds (0.911229x), total internal time 409.438 to 399.152 seconds, inprocessing 165.858 to 161.557 seconds, search/scaffolding 243.580 to 237.595 seconds, and peak RSS 643976 to 604272 KiB. Exact state and proof identity make the 10.286-second total improvement a clean promotion result.

Rank 27 remains 99.152 seconds above the current 300-second milestone and 120.965 seconds behind frozen Kissat. Reaching 300 seconds will not stop the work. The next increment returns to the earliest portable common-state gap: residual Congruence/ITE if a further source owner remains, otherwise exact Vivify throughput, Substitute, BVE, and broad search throughput/quality in that order of causal evidence. Closed Sweep semantics/throughput, directed bridge, BVE antecedent admission, and prior Congruence repairs remain closed absent contradictory evidence. Any future fuzz requirement is delegated entirely to Claude Code under the new §8.2 contract.